Optical Applications in Detecting Microplastics

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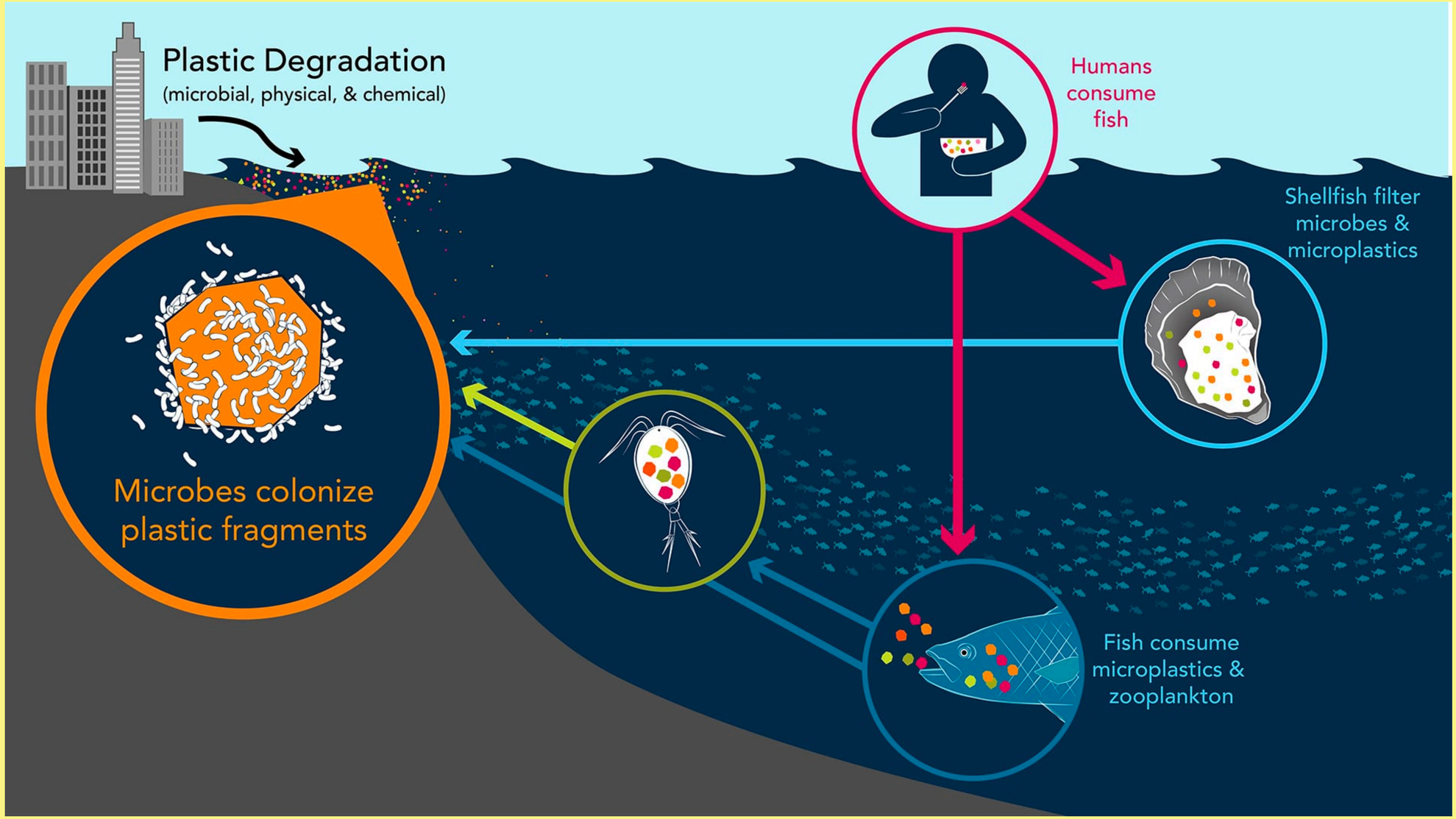
What are microplastics?

- Tiny fragments of plastic
- Between 1mm and 5mm
- Fibres, films, pellets and sponges
- Caused by weathering of plastic waste in water
- Also produced by most manufacturing processes

Why are they bad?



(Natalie Renier, WHOI Creative)
https://www.whoi.edu/oceanus/feature/junk-food/

Why microplastics are hard to detect

- Very small
- Translucent in air, almost transparent in water
- Usually detected (slowly) in a chemistry lab
- It would be nice to have a portable method

https://www.euractiv.com/section/health-consumers/news/who-says-microplastics-in-water-not-a-health-risk-more-research-needed/

How optics can help

- Hyperspectral cameras on drones can give spectral information
- Great for coloured plastics, but not effective for translucent plastics
- New methods use the effects of surface roughness on transmission spectrum
- This can be done using a handheld device, or permanent fixture
- Devices can be placed in rivers, on drones, or on cargo ships
- Some limitations apply

Current limitations and how we might overcome them

- Difficulties powering permanent on-site devices
  - Renewables can feasibly solve this
- Data transmission methods can be unreliable
  - Wireless systems and networks are more commonplace for this
- Individual technologies exist already
  - All that remains is for them to be combined

References

- https://www.optica-opn.org/home/articles/volume_31/november_2020/features/detecting_microplastics_with_optics/
- https://www.whoi.edu/oceanus/feature/junk-food/
- https://www.euractiv.com/section/health-consumers/news/who-says-microplastics-in-water-not-a-health-risk-more-research-needed/

Thank you!

- Questions?